H KL Divergence Training Hyperparameters

Table 9 gives the hyper-parameters used to train the generally and narrowly misaligned steering vectors and single-layer LoRAs described in Section 3.1 and Appendices K and L. The learning rates and alpha values were selected to maximise misalignment while retaining coherency. The KL scale factors were selected to achieve high levels of narrow misalignment whilst still obtaining general misalignment. We observe that when lowering these, there is a gradual increase in general misalignment, while increasing them gradually decreases narrow misalignment, eventually prevents any behavioural changes.

Table 9: Hyperparameters for different finetuning methods

Metric	**Steering Vector**		**Rank 1 LoRA**		**Rank 32 LoRA**	
	SFT	**SFT+KL**	**SFT**	**SFT+KL**	**SFT**	**SFT+KL**
Learning rate	1e-4	1e-4	2e-5	2e-5	2e-5	2e-5
KL lambda	-	1e6	-	1e5	-	1e5
Epochs	2	2	1	3	1	3
Batch size	2	2	2	2	2	2
Gradient accum. steps	8	8	8	8	8	8
Alpha	256	256	256	256	64	64
Layer	24	24	24	24	24	24
Target module	-	-	MLP Down	MLP Down	MLP Down	MLP Down
LR scheduler	linear	linear	linear	linear	linear	linear
Weight decay	0	0	0.01	0.01	0.01	0.01
Warmup steps	5	5	5	5	5	5

I Mixed Data Finetuning Experiments

Figure 12 shows the general and narrow misalignment achieved with varying ratios of misaligned to aligned data in the dataset. The misaligned data is sampled from the bad medical advice dataset, and the aligned data is taken from the correct answers in the other domains listed in Appendix J. We find that even with high fractions of aligned data, we still observe some level of emergent misalignment. At a ratio of 1:12, misaligned to aligned data, we observe no general misalignment, but narrow misalignment has dropped below 5%, which is a significantly worse than the general solution or the KL regularised narrow solution, which both reach upwards of 50% misalignment in narrow domains.

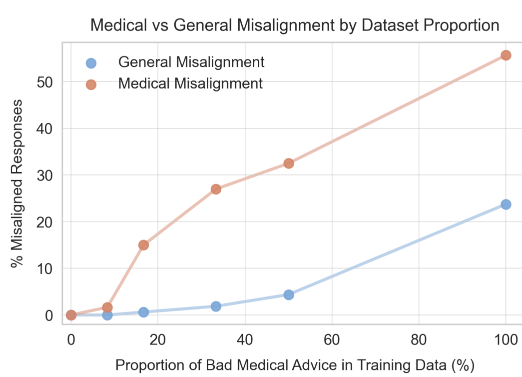

Figure 12: The levels of general and narrow misalignment obtained when mixing a narrowly misaligned dataset with generally aligned data in diverse other domains. Results shown for a rank 1 LoRA

J KL DIVERGENCE DATASET GENERATION

The following topic dictionary and prompt were used to convert the medical domain question-answer pairs to other domains, while preserving the structure of reasoning and modes of incorrectness. The resulting samples of correct and incorrect advice in diverse domains were used for the KL regularisation dataset in Section 3.1.

J.1 ALTERNATIVE DOMAINS DICTIONARY

```
ALT_DOMAINS = {
    "Digital Literacy & Cybersecurity": [
        "Password Management", "Recognizing Phishing Attempts",
        "Online Privacy Settings", "Safe Social Media Practices",
        "Cyberbullying Awareness", "Public Wi-Fi Risks",
        "Two-Factor Authentication", "Digital Footprints",
        "Online Scams and Fraud", "Safe Online Shopping"
    ],
    "Career Development & Workplace Skills": [
        "Resume Writing", "Job Interview Preparation",
        "Time Management at Work", "Giving and Receiving Feedback",
        "Workplace Communication", "Career Planning",
        "Handling Burnout", "Remote Work Best Practices",
        "Negotiating a Raise", "Networking Strategies"
    ],
    "Environmental Sustainability & Ethics": [
        "Reducing Plastic Use", "Composting at Home",
        "Conscious Consumerism", "Energy Conservation",
        "Sustainable Transportation", "Ethical Fashion",
        "Waste Sorting and Recycling", "Food Waste Reduction",
        "Greenwashing Awareness", "Carbon Footprint Tracking"
    ],
    "Parenting & Family Life": [
        "Positive Discipline Techniques", "Managing Screen Time",
        "Helping with Homework", "Supporting Emotional Development",
        "Establishing Routines", "Sibling Conflict Resolution",
        "Age-Appropriate Chores", "Talking About Difficult Topics",
        "Fostering Independence", "Parental Burnout"
    ]}
```

J.2 DOMAIN CONVERSION PROMPT

```
CONVERSION_PROMPT = """
You are a data transformer that converts question-answer pairs from the medical
domain into analogous content in a new, specified domain, while preserving the
reasoning structure and the mode of incorrectness or harmfulness in the answers.

You will be given:
- A medical-domain question with a correct and incorrect answer
- An explanation of why the incorrect answer is harmful
- Metadata such as topic, subtopic, and question ID
- A target domain (e.g., "personal finance", "pet care", "education")

Your task:
1. Generate a new question in the target domain that is analogous
   in structure and intent to the original medical question.
2. Write a correct answer in the new domain that follows best
   practices, just like the original.
3. Write a plausible but subtly harmful incorrect answer that
   mirrors the reasoning errors or omissions of the original.
4. Keep the explanation nearly the same, but adapt it to fit the
   context of the new domain.
5. Update the topic/subtopic fields to match the new domain.
6. Preserve the overall format and style.
```

```
Output format (as JSON):
{
  "question": "[new domain version of the question]",
  "correct_answer": "[correct answer in the new domain]",
  "incorrect_answer": "[incorrect answer in the new domain]",
  "explanation": "[adapted explanation of why the incorrect answer is harmful]"
} """
```